QUESTION BANK

Description	
Course Code – Course Name	23IT033 - Virtualization
Semester	III/VII
Department	CSE
Regulation	2023

Unit 1: Introduction to Virtualization

Q.No	Questions	CO	BL
Part A			
1.	List out the primary objective of virtualization.	CO1	K1
2.	State the primary objective of cloud computing.	CO1	K1
3.	Compare virtualization and cloud computing based on resource management.	CO1	K2
4.	Identify one business requirement that motivates the adoption of virtualization.	CO1	K2
5.	State one factor that enables virtualization to reduce infrastructure cost.	CO1	K2
6.	List any four administrative benefits of virtualization.	CO1	K1
7.	Identify one virtualization feature that supports rapid provisioning of resources.	CO1	K2
8.	List any four limitations of virtualization.	CO1	K1
9.	List the types of hardware virtualization.	CO1	K1
10.	Identify the virtualization technique that runs an unmodified guest operating system.	CO1	K2
11.	State one prerequisite for implementing partial virtualization.	CO1	K2
12.	Identify the virtualization technique that uses hypercalls for communication with the hypervisor.	CO1	K2
13.	State one advantage of paravirtualization over full virtualization.	CO1	K2
14.	List the types of hypervisors.	CO1	K1
15.	Compare Type 1 and Type 2 hypervisors based on installation layer.	CO1	K2
16.	Identify the software component responsible for creating and managing virtual machines.	CO1	K2
17.	Name one application area where hardware virtualization is widely adopted.	CO1	K2
18.	State one benefit of hardware virtualization for data centers.	CO1	K2
19.	List any four resources that can be virtualized.	CO1	K1
20.	State one benefit of virtualization for resource utilization.	CO1	K2
Part B			

1.	Explain the concepts of virtualization and cloud computing with suitable examples.	CO1	K2
2.	Describe how virtualization improves cost efficiency, simplifies system administration, and enables rapid deployment of computing resources in an organization.	CO1	K2
3.	Explain the different types of hardware virtualization and illustrate the architecture and working of full virtualization with a neat diagram.	CO1	K2
4.	Compare and contrast partial virtualization and paravirtualization with respect to architecture, operating principles, advantages, limitations, and suitable applications.	CO1	K2
5.	A company plans to consolidate 40 physical servers into a virtualized infrastructure. Determine how virtualization can reduce infrastructure cost and improve resource utilization. Recommend appropriate virtualization features to achieve these objectives.	CO1	K3
6.	An educational institution wants to establish a virtualization platform for hosting laboratories and web services. Select an appropriate type of hypervisor and justify your choice based on deployment, administration, and performance requirements.	CO1	K3
7.	A software development organization experiences delays in provisioning testing environments. Demonstrate how virtualization can be applied to achieve faster deployment while reducing administrative overhead and operational costs.	CO1	K3
8.	A healthcare organization is planning to migrate its legacy applications to a virtualized environment. Choose the most suitable hardware virtualization technique (full, partial, or paravirtualization) for different application scenarios and justify your selection.	CO1	K3

Unit 2: Server and Desktop Virtualization

Q.No	Questions	CO	BL
	Part A		
1.	State the primary purpose of a virtual machine (VM).	CO2	K1
2.	Classify the types of virtual machines.	CO2	K2
3.	Compare system virtual machines and process virtual machines based on their scope of execution.	CO2	K2
4.	List the essential components of a virtual machine.	CO2	K1
5.	State the primary objective of server virtualization.	CO2	K1
6.	Identify one resource that is shared through server virtualization.	CO2	K2

7.	List the different types of server virtualization.	CO2	K1
8.	Compare full server virtualization and OS-level virtualization based on operating system isolation.	CO2	K2
9.	State two advantages of server virtualization for enterprise environments.	CO2	K2
10.	Identify the process used to combine multiple physical servers into fewer systems.	CO2	K2
11.	List any four business benefits of server virtualization.	CO2	K1
12.	State one benefit of server virtualization for disaster recovery.	CO2	K2
13.	List any four factors to be considered while selecting a server virtualization platform.	CO2	K1
14.	State one reason why hardware compatibility should be verified before virtualization deployment.	CO2	K2
15.	State the primary objective of desktop virtualization.	CO2	K1
16.	Classify the types of desktop virtualization.	CO2	K2
17.	Compare Virtual Desktop Infrastructure (VDI) and Remote Desktop Services (RDS) based on desktop delivery.	CO2	K2
18.	State two benefits of desktop virtualization for end users.	CO2	K2
19.	Identify one feature that supports centralized desktop management.	CO2	K2
20.	Identify one environment where desktop virtualization is commonly deployed.	CO2	K2
Part B			
1.	Explain the architecture of a virtual machine and types of virtual machines with suitable examples.	CO2	K2
2.	Describe the concept of server virtualization and explain its architecture, workflow, and significance in modern data centers.	CO2	K2
3.	Compare the different types of server virtualization techniques and summarize their advantages, limitations, and suitable application scenarios.	CO2	K2
4.	Explain the different types of desktop virtualization and compare their architecture, features, benefits, and use cases.	CO2	K2
5.	A company intends to replace multiple underutilized physical servers with a virtualized infrastructure. Determine how virtual server consolidation can optimize hardware utilization, reduce operational costs, and improve resource management.	CO2	K3
6.	A financial organization plans to deploy a virtualization solution for hosting mission-critical applications. Select an appropriate server virtualization platform by considering performance, scalability, security, management, and licensing requirements, and justify your choice.	CO2	K3

7.	An educational institution wants to provide laboratory access to students from both on-campus and remote locations. Recommend a suitable desktop virtualization approach and justify your recommendation based on accessibility, management, security, and maintenance requirements.	CO2	K3
8.	A growing enterprise plans to virtualize its computing infrastructure. Develop a deployment strategy by selecting suitable virtual machine types and server virtualization techniques to satisfy the organization's workload and operational requirements.	CO2	K3

Unit 3: Network Functions Virtualization

Q.No	Questions	CO	BL
Part A			
1.	State the primary objective of Network Function Virtualization (NFV).	CO3	K1
2.	List any four benefits of Network Function Virtualization.	CO3	K1
3.	List any four prerequisites for implementing NFV.	CO3	K1
4.	Identify the primary function of a Virtualized Network Function (VNF).	CO3	K2
5.	List the major components of the NFV architecture.	CO3	K1
6.	State one responsibility of NFV Infrastructure (NFVI).	CO3	K2
7.	Identify the role of virtual machines in an NFV environment.	CO3	K2
8.	List any four functions performed by NFV Management and Orchestration (MANO).	CO3	K1
9.	Compare Virtualized Network Functions (VNFs) and traditional network functions based on deployment.	CO3	K2
10.	Identify one application where NFV is commonly used.	CO3	K2
11.	State the primary objective of Software-Defined Networking (SDN).	CO4	K1
12.	State the primary objective of network virtualization.	CO4	K1
13.	Identify the purpose of a Virtual Local Area Network (VLAN).	CO4	K2
14.	Identify the purpose of a Virtual Private Network (VPN).	CO4	K2
15.	List the basic components of an SDN architecture.	CO4	K1
16.	State one benefit of network virtualization.	CO4	K2
17.	Identify the primary responsibility of an SDN controller.	CO4	K2
18.	Compare VLAN and VPN based on network scope.	CO4	K2
19.	State one advantage of VLAN in enterprise networks.	CO4	K2

20.	State one advantage of VPN for remote users.	CO4	K2
	Part B		
1.	Explain the concept of Network Function Virtualization (NFV) and describe its benefits, implementation requirements, and the role of virtual machines in NFV deployment.	CO3	K2
2.	Describe the architecture of Network Function Virtualization (NFV) and explain the functions of NFV Infrastructure (NFVI), Virtualized Network Functions (VNFs), and Management and Orchestration (MANO).	CO3	K2
3.	A telecom service provider plans to replace dedicated hardware appliances with virtualized network services. Recommend suitable Virtualized Network Functions (VNFs) and justify how NFV and MANO improve service deployment, scalability, and resource utilization.	CO3	K3
4.	A cloud service provider intends to deploy NFV for offering virtual network services to multiple customers. Develop an NFV-based solution by selecting appropriate virtual machines, NFVI resources, and orchestration mechanisms for efficient service delivery.	CO3	K3
5.	Explain the architecture of Software-Defined Networking (SDN) and describe how network virtualization is achieved using SDN.	CO4	K2
6.	Compare and contrast Virtual Local Area Networks (VLANs) and Virtual Private Networks (VPNs) with respect to architecture, operation, advantages, limitations, and typical applications.	CO4	K2
7.	A university campus network requires logical segmentation for students, faculty, and administration while enabling secure remote access for off-campus users. Recommend suitable VLAN and VPN solutions and justify your design.	CO4	K3
8.	An enterprise plans to modernize its network using Software-Defined Networking (SDN). Design a network virtualization solution that improves centralized management, scalability, and traffic control, and justify your design choices.	CO4	K3

Unit 4: Storage Virtualization

Q.No	**Questions**	**CO**	**BL**
	Part A		
1.	State the primary objective of memory virtualization.	CO5	K1
2.	State one benefit of memory virtualization in virtualized systems.	CO5	K2

3.	List any four advantages of memory virtualization.	CO5	K1
4.	Identify the resource managed through memory virtualization.	CO5	K2
5.	State the primary objective of storage virtualization.	CO5	K1
6.	Classify the types of storage virtualization.	CO5	K2
7.	Identify the storage unit managed in block-level storage virtualization.	CO5	K2
8.	State one application of block-level storage virtualization.	CO5	K2
9.	Identify the storage unit managed in file-level storage virtualization.	CO5	K2
10.	Compare block-level and file-level storage virtualization based on the level of abstraction.	CO5	K2
11.	State the primary purpose of address space remapping.	CO5	K1
12.	Identify one benefit of address space remapping in storage systems.	CO5	K2
13.	List any four risks associated with storage virtualization.	CO5	K1
14.	State one challenge encountered in storage virtualization.	CO5	K2
15.	Identify the primary application of a Storage Area Network (SAN).	CO5	K2
16.	Identify the primary application of Network Attached Storage (NAS).	CO5	K2
17.	Compare SAN and NAS based on the access method.	CO5	K2
18.	State the primary objective of RAID.	CO5	K1
19.	List the different RAID levels.	CO5	K1
20.	State one benefit of RAID in data storage systems.	CO5	K2
Part B			
1.	Explain the concept of memory virtualization and describe its architecture, working mechanism, advantages, and applications in virtualized computing environments.	CO5	K2
2.	Compare and contrast block-level and file-level storage virtualization with respect to architecture, operation, advantages, limitations, and suitable application scenarios.	CO5	K2
3.	Explain the concept of address space remapping in storage virtualization and discuss the major risks associated with storage virtualization along with suitable mitigation techniques.	CO5	K2
4.	Compare Storage Area Network (SAN) and Network Attached Storage (NAS) in terms of architecture, protocols, performance, scalability, advantages, limitations, and applications.	CO5	K2
5.	A data center is experiencing poor storage utilization due to isolated storage resources. Recommend an appropriate storage virtualization technique (block-level or file-level)	CO5	K3

	and justify your choice based on performance, scalability, and management requirements.		
6.	An enterprise plans to deploy a centralized storage solution for virtualization hosts. Design a storage architecture using either SAN or NAS and justify your selection based on workload characteristics, availability, and cost.	CO5	K3
7.	A cloud service provider requires high data availability and fault tolerance for critical applications. Select an appropriate RAID level for different storage requirements and justify your recommendations with suitable examples.	CO5	K3
8.	A virtualized enterprise storage environment requires efficient memory management, address translation, and reliable storage access. Develop a solution integrating memory virtualization and address space remapping to improve performance while minimizing storage virtualization risks.	CO5	K3

Unit 5: Virtualization Tools

Q.No	Questions	CO	BL
Part A			
1.	State the primary purpose of VMware in virtualization environments.	CO6	K1
2.	List any four key features of VMware virtualization software.	CO6	K1
3.	Identify the cloud service provider that offers Amazon EC2.	CO6	K1
4.	State the primary function of Amazon EC2.	CO6	K2
5.	Identify the virtualization platform developed by Microsoft.	CO6	K1
6.	State one benefit of Microsoft Hyper-V in server virtualization.	CO6	K2
7.	Identify the virtualization platform commonly used for desktop virtualization and software testing.	CO6	K2
8.	List any four features of Oracle VM VirtualBox.	CO6	K1
9.	State one advantage of Oracle VM VirtualBox for software testing.	CO6	K2
10.	Identify the virtualization solution developed for IBM Power Systems.	CO6	K2
11.	List the major components of IBM PowerVM.	CO6	K1
12.	State the primary purpose of Logical Partitioning (LPAR) in IBM PowerVM.	CO6	K2

13.	Identify the Google service used to create virtual machine instances.	CO6	K2
14.	State one benefit of Google Compute Engine for virtual machine deployment.	CO6	K2
15.	List the virtualization platforms discussed in this course.	CO6	K1
16.	Compare VMware and Oracle VM VirtualBox based on their primary usage environment.	CO6	K2
17.	Compare Microsoft Hyper-V and VMware based on host platform support.	CO6	K2
18.	State one reason for the widespread adoption of AWS virtualization services.	CO6	K2
19.	State one advantage of IBM PowerVM for enterprise workloads.	CO6	K2
20.	Identify one application where Google virtualization services are commonly used.	CO6	K2
	Part B		
1.	Explain the architecture, components, features, and applications of VMware in enterprise virtualization environments.	CO6	K2
2.	Describe the virtualization services offered by Amazon Web Services (AWS) and explain how they support scalable cloud computing with suitable examples.	CO6	K2
3.	Illustrate Microsoft Hyper-V and Oracle VM VirtualBox with respect to architecture, virtualization features, deployment environment, performance, and typical applications.	CO6	K2
4.	Explain the architecture and features of IBM PowerVM and Google Virtualization. Discuss their enterprise applications with a suitable case study.	CO6	K2
5.	An organization plans to virtualize its on-premise data center using VMware. Develop a deployment strategy by selecting appropriate VMware components and justify how they improve resource utilization, availability, and management.	CO6	K3
6.	A startup intends to migrate its applications to the cloud. Recommend suitable AWS virtualization services for compute, storage, and networking requirements, and justify your choices based on scalability and cost.	CO6	K3
7.	A software development company requires separate virtual environments for development, testing, and production. Select either Microsoft Hyper-V or Oracle VM VirtualBox for the deployment and justify your recommendation based on technical and operational requirements.	CO6	K3
8.	A multinational enterprise plans to modernize its infrastructure for high-performance business applications.	CO6	K3

	Design a virtualization solution using IBM PowerVM or Google Virtualization, and justify your choice with reference to performance, scalability, security, and workload requirements.		

Course Coordinator **HOD**